

CT1 Question Paper
PART A (Any 2) (2 x 20 = 40 marks)

Question 1

- a) Explain rate monotonic scheduling (RMS) with its assumptions. *(5 Marks)*
- b) Explain earliest deadline first (EDF) scheduling and its advantages. *(5 Marks)*
- c) Compare preemptive and non-preemptive scheduling in real-time systems. *(5 Marks)*
- d) Define task, process, and thread in the context of real-time systems. *(5 Marks)*

Question 2

- a) Explain peripheral interfacing with suitable examples. *(5 Marks)*
- b) Elucidate SPI, I2C & UART protocol and its applications. *(15 Marks)*

Question 3

- a) Explain TCP/IP stack with reference to embedded systems. *(10 Marks)*
- b) Discuss the role of bus structure. *(10 Marks)*

CT2 Question Paper
PART A (Any 2) (2 x 20 = 40 marks)

Question 1

- a) How are Petri Nets used to model "Deadlock" and "Conflict" in embedded systems? *(10 marks)*
- b) Describe Specification and Description Language (SDL). *(10 marks)*

Question 2

- a) Distinguish the Hybrid Modeling: Continuous vs. Discrete Dynamics and State Machine Design for Safety *(10 marks)*
- b) Elaborate on Embedded Security & Threat Modeling, Reliability Testing *(10 marks)*

Question 3

- a) Debugging: On-board vs. Task-Level with examples. *(10 marks)*
- b) Compare and contrast Class Diagrams and Component Diagrams. *(10 marks)*

Question 4

- a) Compare Simulation: High-Level vs. Low-Level *(10 marks)*
- b) Explain in detail the UML Behavioral Modelling: Use-Case, Activity Diagrams. *(10 marks)*